Project Abstract

Analog Electronics Project

Project Title

Design of a 1.2V Two-Stage Miller-Compensated CMOS Operational Amplifier using g_m/I_D Methodology in 90nm Technology

Objectives of the Project

To design, characterize, and simulate a stable two-stage Miller-compensated operational amplifier at the transistor level using a 90nm CMOS process. Due to short-channel effects at this technology node, we aim to abandon traditional square-law calculations in favor of the g_m/I_D sizing methodology. The objective is to achieve a balanced trade-off between open-loop gain, bandwidth, and power consumption while ensuring strict phase margin requirements for stability.

Design Specifications

- **Technology Node:** 90nm CMOS (GPDK090)
- **Power Supply (V_{DD}):** 1.2 V
- **Open-Loop DC Gain:** ≥ 50 dB
- **Unity Gain Bandwidth (UGB):** ≥ 20 MHz
- **Phase Margin:** ≥ 60°
- **Slew Rate:** ≥ 10 V/μs
- **Load Capacitance (C_L):** 1 pF
- **Power Dissipation:** ≤ 1 mW

Design Methodology/Approach

1. **Device Characterization:** Generate look-up tables and plots (e.g., g_m/I_D vs. I_D/W, g_m/I_D vs. $g_m r_o$) for single NMOS and PMOS devices in Cadence Virtuoso to map actual physical behavior.
2. **Architecture Selection:** Utilize a standard two-stage topology with a differential input pair, active current mirror load, and a common-source second stage.
3. **Transistor Sizing:** Calculate the required tail current based on the target Slew Rate and load capacitance. Use the extracted g_m/I_D charts to determine the exact channel widths (W) required for the gain and bandwidth targets while operating in moderate to-strong inversion.

4. **Compensation Design:** Size the Miller capacitor (C_c) to split the poles and introduce a nulling resistor (R_z) if necessary to cancel the right-half-plane zero, ensuring a phase margin of at least 60°.

5. **Simulation & Verification:** Perform DC, AC, and Transient analyses using Cadence ADE to verify all specifications are met, iterating on transistor geometries as needed.

EDA Tools/Software to be Used

• Cadence Virtuoso (Schematic Editor, ADE L/XL for simulation)

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